SAMPLE DOCUMENT - for testing UniLife Copilot

Faculty of Computing and Information Technology
BACS2063 Data Structures and Algorithms
Academic Session 2026/2027 | Individual Assignment (20%)

Assignment 1: Library Book Queue System

Submission deadline: Friday, 30 October 2026, 11:59 PM

Overview
Build a console program in Java that manages a waiting queue for popular library books. Students reserve a book; when it is returned, the next student in the queue is notified.

Requirements
1. Implement your own Queue ADT using a linked list (do not use java.util.Queue).
2. Support: reserve a book, cancel a reservation, return a book, show the queue for a book.
3. Store at least 10 sample books and 15 sample students.
4. Handle invalid input without crashing.

Deliverables
- Source code (zipped) uploaded to the e-learning portal.
- A short report (max 6 pages) with your ADT design, class diagram and test screenshots.

Marking
ADT implementation 40%, program features 30%, report 20%, code quality 10%.

Late submission: 10% deducted per day. Estimated effort: about 10 hours.